Question

X, **Y**, and **Z** are three elements with sequentially increasing atomic numbers, among which **X** and **Y** are third-period elements, and only **Y** is a nonmetallic element. **X** and **Z** belong to the same main group, and the number of electrons in the M shell and N shell of the ground-state **Z** atom is the same. The simple substances of both **X** and **Z** can react violently with water to release gas **A**. When the simple substance of **Y** is added to the solution obtained from the reaction, gas **A** is still produced.

A cubic crystal system contains only the elements **X**, **Y**, and **Z**, where **Y** atoms are stacked in a specific manner, forming only pentagonal dodecahedral cages and hexadecahedral cages in a ratio of 2:1. These cages are tightly connected solely through face-sharing. **X** atoms occupy all dodecahedral cages, **Z** atoms occupy all hexadecahedral cages, and each **Y** atom is connected to four other **Y** atoms. Focusing on a single hexadecahedral cage, it has 4 hexagonal faces and 12 pentagonal faces, with four threefold rotation axes.

Which of the following statements is correct:

A. The sixteen-hedral cages of the crystal are not interconnected with each other.

B. A conventional unit cell contains 4 **Z** atoms

C. The simplest integer ratio chemical formula of this crystal contains 18 atoms

D. The simplest integer ratio chemical formula of this crystal contains 4 atoms of the metallic element.

E. It is known that the conventional unit cell of this crystal has $a = 1475.6\ \mathrm{pm}$, and the crystal density is $2.714\ \mathrm{g/cm^3}$.

F. None of the above options is correct

Answer

Correct Answer: E

Detailed Explanation

The number of electrons in the M and N shells of the ground-state atom **Z** is the same, indicating that **Z** is located in the fifth period or later. Both the elemental forms of **X** and **Z** can react violently with water to release gas **A**, suggesting that **X** and **Z** are alkali metals or alkaline earth metals, and **A** is hydrogen.

X and **Y** are third-period elements. **X** can only be **Na** or **Mg**, but **Mg** does not react violently with water, so **X** should be **Na**, and **Z** is **Cs** (radioactive elements are generally not considered). In this case, the reaction system with water forms an alkaline sodium hydroxide solution, and the addition of elemental **Y** still produces hydrogen gas, so **Y** can only be **Si**.

CHECKPOINT 1 PTS

Z is located in the fifth period or later

CHECKPOINT 1 PTS

X,Z are alkali metals or alkaline earth metals, and **A** is hydrogen

CHECKPOINT 2 PTS

X is **Na**, **Z** is **Cs**, and **Y** is **Si**

If the hexadecahedra are not connected by sharing faces, then the hexadecahedra and dodecahedra can only be connected through pentagonal faces. This would result in the hexagonal faces of the hexadecahedra being isolated, and the cubic symmetry would inevitably cause multiple hexagonal faces to form new cages, which contradicts the given conditions. Therefore, the hexadecahedra must be connected by sharing faces, making option A incorrect.

CHECKPOINT 1 PTS

The hexadecahedra must be connected by sharing faces

The hexadecahedral cages possess a threefold rotation axis. For the cubic system, the hexagonal faces of the hexadecahedral cages must be perpendicular to the threefold rotation axis in the conventional unit cell. Based on symmetry, it can be inferred that the hexadecahedral cages must lie along the body diagonals of the conventional unit cell. Thus, the hexadecahedral cages can only occupy the vertex-body center equivalent positions or the equivalent positions at (1/4,1/4,1/4). However, since the hexadecahedral cages share faces, the vertex-body center equivalent positions cannot fully share faces (half the length of the body diagonal is not equal to the edge length). Therefore, in the conventional cubic unit cell, the hexadecahedral cages are located at the equivalent positions of (1/4,1/4,1/4), totaling 8 cages, meaning there are 8 **Z** atoms per conventional unit cell. Option B is incorrect.

CHECKPOINT 2 PTS

The hexadecahedral cages must lie along the body diagonals of the conventional unit cell

CHECKPOINT 2 PTS

In the conventional cubic unit cell, the hexadecahedral cages are located at the equivalent positions of (1/4,1/4,1/4)

CHECKPOINT 1 PTS

There are 8 **Z** atoms per conventional unit cell

Since the ratio of dodecahedral cages to hexadecahedral cages is 2:1, and there are 8 **Cs** atoms per unit cell, there must be 16 **Na** atoms per unit cell.

The number of vertices in a hexadecahedral cage:

$$n_{16} = (\ 5 \times 12 + 6 \times 4) \div 3 = 28$$

The number of vertices in a dodecahedral cage: $n_{12} = 20$

Each silicon atom is shared by four cages, so the number of silicon atoms per unit cell:

$$n_{\mathrm{Si}}{=}(20 \times 16 + 28 \times 8) \div 4{=}136$$

Thus, the simplest chemical formula of this crystal is $\mathrm{Na_2CsSi_{17}}$. Options C and D are both incorrect.

CHECKPOINT 2 PTS

The simplest chemical formula of the crystal is $\mathrm{Na_2CsSi_{17}}$

Calculating the crystal density:

$\rho = \mathrm{ZM/N_{Aa}^3} = 2.714\ \mathrm{g/cm^3}$, making option E correct.

CHECKPOINT 1 PTS

$\rho = \mathrm{ZM/N_{Aa}^3} = 2.714\ \mathrm{g/cm^3}$

In conclusion, option E is correct.